of photodiode is exactly inverse to that of an LED. Photodiode works in reverse bias condition. Its circuit symbol is shown in Figure 10.23(a). The direction of arrows indicates that the light is incident on the photodiode.

The device consists of a *p-n* junction semiconductor made of photosensitive material kept safely inside a plastic case as shown in Figure 10.23(b). It has a small transparent window that allows light to be incident on the *p-n* junction. Photodiodes can generate current when the *p-n* junction is exposed to light and hence are called as light sensors.

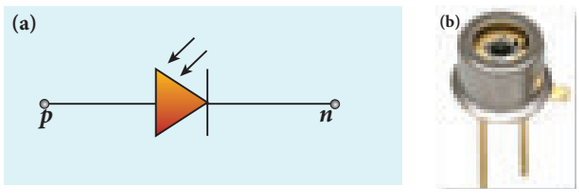


Figure 10.23 (a) Circuit symbol (b) Schematic view of photodiode

When a photon of sufficient energy (hv) strikes the depletion region of the diode, some of the valence band electrons are elevated into conduction band, in turn holes are developed in the valence band. This creates electron-hole pairs. The amount of electron-hole pairs generated depends on the intensity of light incident on the *p-n* junction.

These electrons and holes are swept across the *p-n* junction by the electric field created by reverse voltage before recombination takes place. Thus, holes move towards the *n*-side and electrons towards the *p*-side. When the external circuit is made, the electrons flow through the external circuit and constitute the photocurrent.

When there is no incident light, there exists a reverse current which is negligible. This reverse current in the absence of any incident light is called dark current and is due to the thermally generated minority carriers.

Applications

The photodiodes are used in

- alarm system
- count items on a conveyor belt
- photoconductors
- compact disc players, smoke detectors
- medical applications such as detectors for computed tomography etc.

iii) Solar cell

A solar cell, also known as photovoltaic cell, works on the principle of **photovoltaic effect**. Accordingly, the *p-n* junction of the solar cell generates emf when solar radiation falls on it. The construction details and cross-sectional view are shown in Figure 10.24.

In a solar cell, electron–hole pairs are generated due to the absorption of light photons near the junction. Then the charge carriers are separated due to the electric field of the depletion region. Electrons move towards *n*–type silicon layer and holes move towards *p*-type silicon layer. The electrons reaching the *n*-side are collected by the front contact (metal finger contact) and holes reaching *p*-side are collected by the back

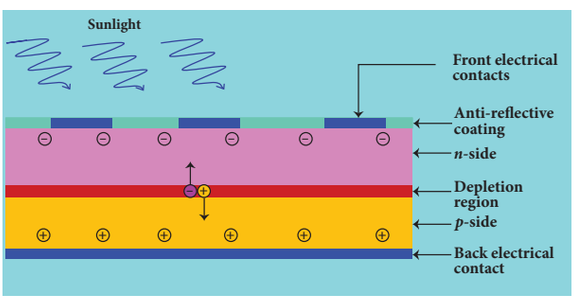


Figure 10.24 Cross-sectional view of a solar cell

UNIT-10_XII-Physics_Vol-2_New.indd 214 21-07-2020 15:35:23